

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- MCQ TEST

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO3	Assessment:	ASSESSMENT TOOL 2- MCQ TEST
Weightage:	25%	Total Marks:	25
CO3 Statement:	<i>Design processor organization by constructing pipelined datapath structures, identifying hazard types (data, control, structural), and comparing superscalar techniques such as out-of-order execution, speculative execution, and simultaneous multithreading (SMT). (BL6)</i>		
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Department:	IT	Date:	30/07/26

ANNEXURE A

1. A processor transfers data between registers before performing an arithmetic operation in the ALU. Which component primarily controls these register transfer operations?

- a) Arithmetic Logic Unit
- b) Control Unit
- c) Memory Unit
- d) Cache Controller

ANS: b) Control Unit

2. In a system where multiple registers are used for intermediate storage, data must be efficiently moved before computation. Which concept ensures correct sequencing of such operations?

- a) Register transfer language
- b) Instruction pipelining
- c) Cache mapping
- d) Interrupt handling

ANS: a) Register transfer language

3. A processor performs logical AND, OR, and XOR operations on data stored in registers. These operations are handled by which unit?

- a) Control Unit
- b) ALU
- c) Bus Interface Unit
- d) Memory Management Unit

ANS: b) ALU

4. A system designer chooses a single-bus architecture for simplicity, but performance issues arise when multiple components attempt data transfer simultaneously. What is the primary limitation here?

- a) Lack of memory
- b) Bus contention
- c) High clock speed
- d) Complex control logic

ANS: b) Bus contention

5. In a multi-bus organization, data transfer between CPU and memory is faster compared to a single-bus system. What is the main reason for this improvement?

- a) Reduced instruction size
- b) Parallel data transfer capability
- c) Increased clock frequency
- d) Simplified hardware

ANS: b) Parallel data transfer capability

6. A processor executes instructions using a 5-stage pipeline. If one stage takes significantly longer than others, what will be the impact?

- a) Increased throughput
- b) Pipeline stalls
- c) Reduced instruction size
- d) Faster execution

ANS: b) Pipeline stalls

7. During pipelined execution, an instruction depends on the result of a previous instruction that has not yet completed. What type of hazard does this represent?

- a) Structural hazard
- b) Control hazard
- c) Data hazard
- d) Memory hazard

ANS: c) Data hazard

8. A pipeline encounters delays due to branching instructions, causing incorrect instruction fetching. Which hazard is this?

- a) Data hazard
- b) Control hazard
- c) Structural hazard
- d) Timing hazard

ANS: b) Control hazard

9. In a system where multiple instructions require the same hardware resource simultaneously, execution is delayed. This is an example of:

- a) Data hazard
- b) Structural hazard
- c) Control hazard
- d) Logical hazard

ANS: b) Structural hazard

10. To reduce pipeline stalls caused by data dependencies, a processor implements forwarding. What does forwarding achieve?

- a) Reduces instruction size
- b) Bypasses intermediate storage
- c) Increases clock speed
- d) Eliminates memory access

ANS: b) Bypasses intermediate storage

11. A superscalar processor issues multiple instructions per clock cycle. What enables this capability?

- a) Single execution unit
- b) Multiple execution units
- c) Larger memory
- d) Slower clock

ANS: b) Multiple execution units

12. In out-of-order execution, instructions are executed in a different order than they appear in the program. What is the main advantage of this technique?

- a) Reduced memory usage
- b) Better utilization of CPU resources
- c) Simplified instruction set
- d) Reduced instruction count

ANS: b) Better utilization of CPU resources

13. A processor predicts the outcome of a branch instruction and executes instructions ahead of time. This technique is known as:

- a) Pipelining
- b) Speculative execution
- c) Multiprocessing
- d) Interrupt handling

ANS: b) Speculative execution

14. In speculative execution, incorrect predictions may lead to discarded results. What is the main drawback of this approach?

- a) Increased memory size
- b) Wasted computation
- c) Reduced clock speed
- d) Increased instruction size

ANS: b) Wasted computation

15. A processor uses separate units to dispatch instructions to execution units. What is the primary role of the dispatch unit?

- a) Execute instructions
- b) Fetch instructions
- c) Assign instructions to execution units
- d) Store instructions

ANS: c) Assign instructions to execution units

16. Execution units in a processor handle different types of operations simultaneously. Which of the following is an example of an execution unit?

- a) Cache memory
- b) ALU
- c) Control Unit
- d) Register file

ANS: b) ALU

17. A processor with hyper-threading allows multiple threads to share execution resources. What is the primary benefit?

- a) Reduced hardware cost
- b) Increased instruction throughput
- c) Lower power consumption
- d) Smaller memory

Ans: b) Increased instruction throughput

18. In simultaneous multithreading (SMT), multiple threads are executed at the same time. What challenge arises from this approach?

- a) Reduced instruction set
- b) Resource contention
- c) Lower clock speed
- d) Reduced memory

Ans: b) Resource contention

19. A system experiences reduced performance when multiple threads compete for the same execution units. This issue is related to:

- a) Instruction pipelining
- b) Resource sharing
- c) Cache coherence
- d) Memory hierarchy

Ans: b) Resource sharing

20. A processor pipeline achieves ideal performance when all stages take equal time. What happens if stages are unbalanced?

- a) Improved speed
- b) Reduced efficiency
- c) Increased memory
- d) No effect

Ans: b) Reduced efficiency

21. In a multi-bus system, data can be transferred between different components simultaneously. What is the main advantage of this design?

- a) Simplicity
- b) Parallelism
- c) Lower cost
- d) Reduced memory

Ans: b) Parallelism

22. A processor encounters frequent stalls due to instruction dependencies. Which technique can best improve performance?

- a) Increasing memory size
- b) Data forwarding
- c) Reducing clock speed
- d) Decreasing instruction count

Ans: b) Data forwarding

23. A superscalar processor executes instructions out of order but commits them in order. Why is this necessary?

- a) To reduce memory usage
- b) To maintain program correctness
- c) To increase clock speed
- d) To simplify hardware

ANS: b) To maintain program correctness

24. A processor uses speculative execution along with branch prediction. What determines the effectiveness of this approach?

- a) Memory size
- b) Prediction accuracy
- c) Instruction size
- d) Number of registers

ANS: b) Prediction accuracy

25. A computing system integrates pipelining, superscalar execution, and SMT. What is the overall goal of combining these techniques?

- a) Reduce hardware complexity
- b) Increase instruction throughput
- c) Decrease memory usage
- d) Simplify programming

ANS: b) Increase instruction throughput

Code of Conduct:

I SREYA.B(192521442)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SREYA.B

MARKS ALLOCATION

	Understanding of Concepts (25%)	Accuracy of Answers (30%)	Application of Knowledge (20%)	Time Management (15%)	Logical Thinking and Decision Making (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginner
Understanding of Concepts	25%	Demonstrates clear and complete understanding of concepts tested in MCQs	Shows good understanding of concepts with minor conceptual errors	Partial understanding; misses key concepts	Misinterprets or fails to understand concepts
Accuracy of Answers	30%	Selects correct answers for all questions with high accuracy	Answers most questions correctly with few errors	Provides limited correct answers with frequent errors	Unable to select correct answers consistently
Application of Knowledge	20%	Applies concepts effectively to application-based and analytical questions	Applies concepts with some errors	Limited application of concepts; requires improvement	Unable to apply relevant knowledge
Time Management	15%	Completes all questions within allotted time effectively	Completes most questions within time	Attempts questions but struggles with time management	Unable to complete test on time
Logical Thinking and Decision Making	10%	Analyzes options carefully and chooses the most appropriate answer	Makes correct choices based on reasonable analysis	Shows limited reasoning in selecting answers	Answers without reasoning